Tutorial 1: ML Foundations and Mathematical Review

Three required tasks and one optional challenge

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Tutorial 1: from concepts to practice

This tutorial applies the main ideas from Week 1 through calculation, reasoning, and interpretation.

- **ML evaluation:** identify problems in an experimental pipeline.
- **Probability:** use Bayes' rule to reason about rare events.
- **Linear algebra:** study how transformations affect means and covariance.
- **Connection:** relate covariance, eigenvectors, PCA, and SVD.

The goal is to move from knowing the formulas to understanding when, why, and how to use them.

Task 1: audit an ML pipeline

Experimental reasoning ▪ 12 minutes

A medical dataset contains repeated visits from the same patients. The goal is to predict disease status for **previously unseen patients**. About 5% of patients have the disease. A team builds the following pipeline:

1. Randomly split all visits into training and test sets.
2. Compute feature means and standard deviations using all available data.
3. Train several classifiers and select the one with the best test performance.
4. Report the resulting test accuracy.

Questions

1. Identify the problems in this pipeline.
2. Propose a corrected experimental workflow.
3. If missing a diseased patient is especially costly, how should this affect evaluation?

Task 2: Bayes' rule under a rare event

Calculation and interpretation ▪ 12 minutes

Let D denote having a disease and $+$ denote a positive test.

$$\mathbb{P}(D) = 0.02, \qquad \mathbb{P}(+ \mid D) = 0.90, \qquad \mathbb{P}(+ \mid \neg D) = 0.08.$$

Questions

1. Compute $\mathbb{P}(D \mid +)$.
2. For 10,000 people, estimate true positives and false positives.
3. Why can $\mathbb{P}(+ \mid D)$ and $\mathbb{P}(D \mid +)$ be very different?

Task 3: how does a linear transformation change covariance?

Derivation and interpretation ▪ 18 minutes

Let

$$\mathbb{E}[\boldsymbol{X}] = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \boldsymbol{\Sigma}_X = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}.$$

Define

$$\boldsymbol{Y} = \boldsymbol{A}\boldsymbol{X} + \boldsymbol{b}, \qquad \boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Questions

Derive $\mathbb{E}[\boldsymbol{Y}]$ and $\boldsymbol{\Sigma}_Y$. Then interpret the variances and covariance of Y_1 and Y_2.

Optional challenge: from covariance to PCA

Consider

$$\boldsymbol{\Sigma} = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}.$$

1. Find its eigenvalues and unit eigenvectors.
2. Which direction has the largest variance?
3. Which direction would PCA keep first?

Connection to SVD

If a centered data matrix satisfies $\boldsymbol{X} = \boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^{\mathsf{T}}$, which matrix contains the PCA directions?

Tutorial 1

Questions and discussion